

Total No. of Questions: 8]

SEAT No. :

P2769

[6002]-286

[Total No. of Pages : 3

S.E. (Computer Engineering/IT)
ENGINEERING MATHEMATICS-III
(2015 Pattern) (Semester-II) (207003)

Time : 2 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.No.1 OR Q.No. 2, Q.No. 3 Or Q.No. 4, Q.5 or Q.No.6, Q.No. 7 Or Q.No. 8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Your answer will be valued as a Whole.
- 4) Figures to the right indicate full marks.
- 5) Assume suitable data, if necessary.

Q1) a) Solve any two of the following: [8]

i) $\frac{d^2y}{dx^2} + y = \tan x$ (use method of variation of parameters)

ii) $(D^2 + 3D + 2)y = e^{4x} + 5x$

iii) $x^2 \frac{d^2y}{dx^2} - 3x \frac{dy}{dx} + 5y = x^2 \sin(\log x)$

b) Solve the following integral equation using Fourier transform [4]

$$\int_0^{\infty} F(x) \sin \lambda x dx = \begin{cases} 1 & 0 \leq \lambda \leq 1 \\ 0 & \lambda \geq 2 \end{cases}$$

OR

Q2) a) An electrical circuit consists of an inductance 0.1 henry, a resistance R of 20 ohms & a condenser of capacitance C of 25 microfarads. If the diff eqⁿ of electric circuit $L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = 0$. Then find the charge q and current i at any time t, given that when $t = 0$, $q = 0.05$ coulombs and $i = \frac{dq}{dt} = 0$. [4]

P.T.O.

b) Solve any one [4]

i) Find $Z^{-1} \left\{ \frac{1}{(z - \frac{1}{2})(z - \frac{1}{3})} \right\}$

ii) Find Z transform of $f(k) = 2^k \sin(3k + 2), k \geq 0$

c) Using Z transform solve the difference eqⁿ [4]

$12f(k+2) - 7f(k+1) + f(x) = 0$ where $f(0) = 0, f(1) = 3, k \geq 0$

Q3) a) The first four moments of a distribution about the value 4 are -2, 0, 1 and 3 obtain the first four central moments β_1 and β_2 [4]

b) Fit a straight line of the form $y = ax + b$ to the following data by the least squares method. [4]

x	-2	-1	1	3	4	6
y	3	4	-1	2	-2	0

c) A die is thrown 8 times and it is required to find the probability that 3 will show [4]

- i) Exacty 2 times
- ii) At least seven times
- iii) at least once

OR

Q4) a) The life of army shoes is normally distributed with mean 8 months and standard deviation 2 months. If 5000 pairs are issued how many pairs would be expected to need replacement after 12 months? [Given : $P(z \geq 2) = 0.0228$] [4]

b) Suppose that a book of 600 pages contains 40 printing mistakes. Assume that these errors are randomly distributed throughout the book and x, the number of errors per page has a poisson distribution. What is the probability that 10 pages selected at random will be free of errors? [4]

c) The regression equations calculated from a given set of observations for two random variables are [4]

$$x = -0.4y + 6.4$$

$$y = -0.6x + 4.6$$

calculate $\bar{x}, \bar{y}$ and ϵ .

- Q5) a)** Find the directional derivative of $\phi = xy^2 + yz^3$ at $(1, -1, 1)$ in the direction of vector $\bar{i} + 2\bar{j} + 2\bar{k}$. [4]
- b)** Show that $\text{curl curl curl } \bar{E} = \nabla^3 \bar{E}$, if $\bar{E}$ is solenoidal vector field. [4]
- c)** Evaluate the line integral of vector point function $\bar{F} = (x^2 - y^2)\bar{i} + 2xy\bar{j}$ along the curve $y^2 = x$ from point $(0, 0)$ to $(1, 1)$ in xy plane. [5]

OR

- Q6) a)** If directional derivative of $\phi = ax^2y + by^2z + cz^2x$ at $(1, 1, 1)$ has maximum magnitude 15 in the direction parallel to line $\frac{x-1}{2} = \frac{y-3}{-2} = \frac{z}{1}$ find a, b, c . [4]
- b)** Show that (any one) [4]
- i) $\nabla(\bar{a} \cdot \bar{r}) = \bar{a}$
- ii) $\nabla \times (\bar{a} \times \bar{r}) = 2\bar{a}$
- c)** Using Green's theorem, evaluate the integral $\oint_C (xydx + y^2dy)$ over the area bounded by curves $y = x^2$ & line $y = x$ in first quadrant. [5]

- Q7) a)** If $F(z) = u + iv$ is analytic, find $f(z)$ if $u + v = e^{-x}(\cos y - \sin y)$. [4]
- b)** Evaluate $\oint_C \frac{e^z}{(z+1)^2(z+2)^2} dz$ where C is $|z+1| = \frac{1}{2}$. [5]
- c)** Show that under transformation $w = \frac{i-z}{i+z}$, x axis in z plane maps on to circle $|w| = 1$. [4]

OR

- Q8) a)** Show that analytic Function with constant modulus is constant. [4]
- b)** Evaluate $\oint_C \frac{2z^2 + 2z + 1}{(z+1)^2(z-3)} dz$ where C is $|z+1| = 2$. [5]
- c)** Find bilinear transformation which maps $-i, 0, 2+i$ of z plane onto $0, -2i, 4$ of w -plane. [4]

